

Approximating Directed Buy-at-Bulk Network Design

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Abstract. Buy-at-bulk network design is a well-known problem that has been researched extensively on undirected graphs. In this paper, we initiate the theoretical study of the same problem on directed graphs, thus capturing real-life situations where the cost of installing capacity on an edge is asymmetric with respect to direction, as e.g. in the design of wireless and satellite communication networks.

More specifically, we develop two approximation algorithms for directed buy-at-bulk network design in the non-uniform cost model. Combined, they achieve a ratio of $O(\min \{k^{1/2+\epsilon} n^{4/5+\epsilon}\})$ for any constant $\epsilon > 0$, where n is the number of nodes of the network and k is the number of demand pairs to be connected. Further, the above ratio is independent of the amount of traffic flow requested by each demand pair, which may vary arbitrarily, and it can be remarkably improved when all demand pairs share a common sink (or a common source, symmetrically).

To the best of our knowledge, this is the first non-trivial approximation factor established for the aforementioned problem, and naturally it also applies to more restricted cost models, such as the uniform and rent-or-buy models. Moreover, it essentially matches the best currently known approximation guarantees for directed Steiner forest, which may be viewed as a special case of directed buy-at-bulk network design.

1 Introduction

Network design has long been a fundamental area of combinatorial optimization. In a typical such problem, we are given a network topology and a list of pairs of network nodes, each with an associated traffic demand. The goal is to determine how to route in the network the traffic demand between each of those pairs, so that some objective function is optimized—such as minimizing the cost of installing the requisite capacity to support all traffic. For more than a decade, researchers have devoted a lot of attention to the buy-at-bulk aspect of network design, in which cost is a sub-additive function of capacity, motivated by the fact that actual costs of network components tend to exhibit economies of scale. However, the theoretical study of the above problem has been limited to undirected graphs so far; by contrast, in this work we focus on directed graphs.

Asymmetric networks. To model a real-life problem more accurately, directed graphs are often indispensable. Perhaps the most prominent such example is the

asymmetric traveling salesman problem: among other things, it is straightforward to represent one-way streets by directed edges, whereas undirected edges would obviously be inappropriate for that purpose. Similar concerns arise in network design when the cost of carrying a unit of traffic across an edge varies substantially depending on direction, as is often the case in communication networks [21,5], e.g. if they contain satellite and/or wireless links. It is worth pointing out that as the popularity of these so-called *asymmetric networks* continues to increase, they have become the object of intense study by the networking community. For instance, see [20,22,15], among many others.

The typical cause of the aforementioned phenomenon is the disparity between the transmitter and receiver capabilities of network equipment, which in turn may be due to various technical, geographical, and even regulatory constraints. As an illustrative example, consider a cellular phone communicating with a base station. It is much easier, by comparison, to achieve some desired transfer rate from the latter to the former than vice versa, primarily because the transmission power of the phone is subject to heavy limitations. Hence, just as the buy-at-bulk framework was developed to reflect the economies of scale prevalent in network costs, a directed graph representation is needed to capture common situations where supporting a given bandwidth capacity in one direction along a network edge is markedly more expensive than in the opposite direction.

Cost models. The most general cost model considered in buy-at-bulk network design is called the *non-uniform* cost model. It stipulates that the cost of supporting $\ell \geq 0$ units of traffic on a given edge e is given by a sub-additive, non-negative, and non-decreasing function $c_e(\ell)$, with $c_e(0) = 0$. On the other hand, the *uniform* cost model adds the restriction that there must exist some function $c(\ell)$ such that for every e in the network we have $c_e(\ell) = C_e c(\ell)$, with $C_e \geq 0$ depending only on e . Moreover, the special case where $c(\ell) = 1$ for all $\ell > 0$ is equivalent to the well-known Steiner forest problem.

Any buy-at-bulk cost function $c_e(\ell)$ may be approximated by the lower envelope of polynomially many linear cost functions [2,9]. Namely, we can replace every edge e with $p(e)$ parallel edges $e_1, e_2, \dots, e_{p(e)}$, such that: (a) for each $i = 1, \dots, p(e)$, $c_{e_i}(\ell) = \sigma(e_i) + \delta(e_i) \cdot \ell$ for $\ell > 0$, where $\sigma(e_i), \delta(e_i)$ are non-negative constants called *cost coefficients*; and (b) $c_e(\ell) \leq \min_{i=1, \dots, p(e)} c_{e_i}(\ell) < 3 \cdot c_e(\ell)$. Note also that specifying each new edge's cost coefficients suffices to describe its cost function. Thus, this alternative representation is (approximately) equivalent to the original one, and we shall use it throughout the rest of the paper.

Related work. As mentioned above, there is a sizable body of research work devoted to buy-at-bulk network design, but only in the undirected setting. One of the earliest approximation algorithm formulations of the problem was due to Salman et al. [23]. Subsequently, Awerbuch and Azar [3] obtained a $O(\log n)$ approximation for uniform buy-at-bulk network design, where n is the number of nodes of the network, using probabilistic embeddings into tree metrics [4,12]. For the *single-sink* case (i.e. when all demand pairs have a common sink, as opposed to the more general *multi-commodity* case that has no such restriction), Guha et al. [16] were the first to achieve an $O(1)$ approximation.

Non-uniform buy-at-bulk network design proved to be considerably more challenging to tackle. For the single-sink case, Meyerson et al. [19] presented a randomized $O(\log n)$ approximation algorithm, later derandomized in [10]; there is also an $O(\log^4 n)$ competitive online algorithm [18]. Charikar and Karagiozova [7] gave an $e^{O(\sqrt{\log n \log \log n})}$ approximation for the multi-commodity case, and a polylogarithmic approximation was obtained fairly recently by Chekuri et al. [9], achieving an $O(\log^4 k)$ ratio, where k is the number of demand pairs.

On the other hand, unlike buy-at-bulk network design, Steiner problems have been studied in the directed setting for quite some time. Zelikovsky [24] first gave an $O(k^\epsilon)$ approximation for directed Steiner tree on acyclic graphs. Charikar et al. [5] obtained a slightly better ratio on arbitrary directed graphs using a somewhat simpler algorithm. Furthermore, they achieved an $\tilde{O}(k^{2/3})$ approximation for directed Steiner forest, which was improved to $O(k^{1/2+\epsilon})$ by Chekuri et al. [8] almost a decade later. Very recently, Feldman et al. [13] provided another approximation guarantee for this problem, namely $O(n^{4/5+\epsilon})$. Since k may range from 1 to $O(n^2)$, the latter two results are mutually incomparable.

Additionally, it should be pointed out that directed Steiner problems are closely connected to group Steiner problems. In particular, group Steiner tree (on undirected graphs) is essentially a special case of directed Steiner tree. The relation between group Steiner forest and directed Steiner forest is analogous. Polylogarithmic approximations have been established for both group Steiner tree [14] (using LP-rounding) and group Steiner forest [8].

Finally, regarding lower bounds, Andrews [1] demonstrated that there is no $\Omega(\log^{1/2-\epsilon} n)$ approximation algorithm for non-uniform buy-at-bulk network design in the undirected setting, unless $\text{NP} = \text{ZPTIME}(n^{\text{polylog } n})$. This result clearly distinguishes buy-at-bulk network design from Steiner forest, which is 2-approximable. For directed Steiner forest, however, Dodis and Khanna [11] showed a hardness factor of $\Omega(2^{\log^{1-\epsilon} n})$ assuming $\text{NP} = \text{TIME}(n^{\text{polylog } n})$, which obviously carries over to directed buy-at-bulk network design as well.

Our results. We present two approximation algorithms for directed buy-at-bulk network design in the non-uniform cost model, which together yield an approximation ratio of $O(\min \{k^{1/2+\epsilon}, n^{4/5+\epsilon}\})$ for any constant $\epsilon > 0$, where n is the number of nodes of the network and k is the number of demand pairs to be connected. Note that the ratio is independent of the amount of traffic flow demanded by each such pair, which may vary arbitrarily, and also that it can be reduced to $O(k^\epsilon)$ (or $O(n^\epsilon)$, equivalently) in the single-sink case.

To the best of our knowledge, this is the first substantial improvement on the trivial approximation guarantee of k for this problem. Of course, the above result also covers the uniform cost model as a special case, and furthermore it matches up to a constant factor the best currently known approximation ratios for directed Steiner forest from [8] and [13].

Overview of techniques. Our algorithms extend those in [8] and [13] for directed Steiner forest. More specifically, to obtain the $O(k^{1/2+\epsilon})$ approximation, we first reduce our problem to what may be described as the “buy-at-bulk

generalization” of group Steiner forest, using the so-called *layering* and *path-splitting* transformations [8]. In turn, that is reduced to the buy-at-bulk generalization of group Steiner tree, following the *bucketing-and-scaling* method pioneered in [9]. We then apply a refined version of the LP-rounding scheme developed in [14] for group Steiner tree, and eventually produce a partial solution that connects at least some of the demand pairs, at a reasonable cost. The algorithm is then applied recursively to any demand pairs that remain unconnected.

For the analysis of the above algorithm, it is crucial to establish first a structural property of low-cost solutions, which we achieve by generalizing Zelikovsky’s tree height-reduction procedure [24,17] in a suitable fashion. In particular, the original height-reduction procedure outputs a tree contained in the metric closure of the network graph. In the buy-at-bulk setting, then, we need to define the *bi-metric closure* of a graph, which extends the standard concept of metric closure and may also be of independent interest. We also take advantage of the fact that Zelikovsky’s procedure preserves the lengths of all paths from the root of the tree to its leaves, under *any* metric. This property is usually overlooked, because it is not necessary for typical applications of the procedure to Steiner problems. However, it does play a pivotal role within the bi-metric closure framework that we introduce, since more than one metrics are involved.

On the other hand, to derive the $O(n^{4/5+\epsilon})$ approximation ratio, we partition the demand pairs into two classes, namely the *good* and the *bad*. The good pairs are distinguished by the fact that they can all be connected in a relatively straightforward way, at a reasonable cost. Nevertheless, the situation becomes more complicated for the bad pairs. For these, we have to try out a number of options that include invoking the $O(k^{1/2+\epsilon})$ -approximation algorithm mentioned earlier, and rounding a near-optimal solution to a restricted LP relaxation of the problem. This approach resembles the one employed in [13]; however, the definition of good pairs is more specialized in our context, thus requiring a more elaborate case analysis to handle the bad pairs.

Organization. The next section presents several key concepts, including the formal definition of the directed buy-at-bulk network design problem. In Section 3, we show the structural result that is vital to the analysis of our algorithms, which are then described in Section 4. Lastly, Section 5 discusses some conclusions that may be drawn from our results and suggests directions for future research.

2 Preliminaries

An instance of the directed buy-at-bulk network design problem has the form (G, D) , where $G = (V, E)$ is a directed multigraph with two non-negative cost coefficients $\sigma(e), \delta(e)$ specified for every edge $e \in E$, and $D = \{d_{uv}\}$ is a $|V| \times |V|$ matrix with non-negative integer elements. An ordered pair $(u, v) \in V \times V$ for which $d_{uv} \geq 1$ is called a *demand pair*; furthermore, u and v are the *source* and *sink* of the demand pair, respectively. For simplicity of notation, we denote by $\mathcal{D} \subseteq V \times V$ the set of demand pairs, by k the cardinality of $\mathcal{D}$, and by n the cardinality of V . Additionally, for any path P , let $\sigma(P) = \sum_{e \in P} \sigma(e)$ and $\delta(P) = \sum_{e \in P} \delta(e)$.

A feasible solution to this instance consists of a collection $\mathcal{P} = \{P(u, v)\}$ of k paths in G such that for each demand pair $(u, v) \in \mathcal{D}$, $\mathcal{P}$ contains a path $P(u, v)$ from u to v . We say that $P(u, v)$ *serves* the demand pair (u, v) in the solution $\mathcal{P}$. The goal is to minimize the *cost* of $\mathcal{P}$:

$$c(\mathcal{P}) = \sum_{e \in \mathcal{P}} \sigma(e) + \sum_{P(u, v) \in \mathcal{P}} \delta(P(u, v)) \cdot d_{uv} = \sum_{e \in \mathcal{P}} (\sigma(e) + \delta(e) \cdot \ell(e)), \quad (1)$$

where the notation $e \in \mathcal{P}$ indicates that edge e belongs to at least one of the paths in $\mathcal{P}$, and $\ell(e)$ is the *load* of edge e and is equal to $\sum d_{uv}$, with the sum taken over all $(u, v) \in \mathcal{D}$ such that $e \in P(u, v)$.

Without loss of generality, we henceforth assume that: (a) $d_{vv} = 0$ for all $v \in V$; (b) if a node of G is the source of a demand pair, then it has in-degree zero; and (c) if a node is the sink of a demand pair, then it has out-degree zero. The latter two properties can be guaranteed easily by a simple transformation, so that the number of nodes of G is at most tripled and the cost of the optimal solution remains unaffected.

A *partial solution* is a collection $\mathcal{P} = \{P(u, v)\}$ of paths that serve a non-empty subset $\mathcal{D}(\mathcal{P})$ of the demand pairs, where again $P(u, v)$ denotes the path serving the pair $(u, v) \in \mathcal{D}(\mathcal{P})$. The cost $c(\mathcal{P})$ of the partial solution $\mathcal{P}$ is given by (1), similarly to the case of a feasible solution. Moreover, the quantity $\text{den}(\mathcal{P}) = c(\mathcal{P})/|\mathcal{D}(\mathcal{P})|$ is called the *density* of the partial solution $\mathcal{P}$. In particular, if $\mathcal{P}$ is a feasible solution then its density is simply $\text{den}(\mathcal{P}) = c(\mathcal{P})/k$.

Finally, we introduce a concept that generalizes the metric closure of a graph.

Definition 1. Given a directed multigraph $G = (V, E)$ with two non-negative cost coefficients $\sigma(e), \delta(e)$ specified for every $e \in E$, we define the bi-metric closure $\bar{G} = (V, \bar{E})$ of G as a directed multigraph such that: (a) for each $(u, v) \in V \times V$, $u \neq v$, and for every distinct (simple) path P from u to v in G , there exists a directed edge $\bar{e} \in \bar{E}$ from u to v with cost coefficients $\sigma(\bar{e}) = \sigma(P)$ and $\delta(\bar{e}) = \delta(P)$; and (b) $\bar{E}$ contains no edges other than those required by (a).

Observe that $\bar{G}$ may be exponential in size. Fortunately, for our purposes we can avoid computing it in its entirety, as we shall see below.

Definition 2. Given an instance (G, D) of directed buy-at-bulk network design, let $\ell_{\max} = \sum_{(u, v) \in \mathcal{D}} d_{uv}$, which is an upper bound of the load on any edge in any feasible solution. We define the restricted bi-metric closure $\hat{G} = (V, \hat{E})$ of G as a subgraph of $\bar{G}$ that is minimal under the following property: for each integer $0 \leq z < \log \ell_{\max}$ and for every $(u, v) \in V \times V$, $u \neq v$, there exists an edge $\hat{e} \in \hat{E}$ from u to v such that for any other $\bar{e} \in \bar{E}$ from u to v , $\sigma(\hat{e}) + \delta(\hat{e}) \cdot 2^z \leq \sigma(\bar{e}) + \delta(\bar{e}) \cdot 2^z$.

Clearly, $\hat{G}$ is computable by executing a standard all-pairs shortest path routine on G with edge weights $\sigma(e) + \delta(e) \cdot z$, for each possible value of z . This takes time $\text{poly}(n, \log \ell_{\max})$ overall, which is polynomial with respect to the size of the instance (G, D) .

3 Structural Results

The goal in this section is to show that there always exist partial solutions of a very specific form that have sufficiently low density.

Definition 3. A junction tree J is the union of an in-branching J_I and an out-branching J_O , both rooted at some node r which is called the junction node. J_I and J_O are not required to be edge-disjoint. The height of J is the maximum of the heights of J_I and J_O , i.e. the maximum length (in terms of the number of edges) of a simple path in J having one endpoint at r .

Furthermore, the *support* of a partial solution $\mathcal{P} = \{P(u, v)\}$ is the union of the paths it contains, i.e. $\bigcup P(u, v)$. The first lemma below, whose proof we omit due to space constraints, is analogous to Lemma 3.1 in [8]. Combining it with the second lemma leads to Corollary 1.

Lemma 1. *There exists a partial solution $\mathcal{P}^\circ = \{P^\circ(u, v)\}$ whose support is a junction tree $J^\circ \subseteq G$ and $\text{den}(\mathcal{P}^\circ) \leq \sqrt{k} \cdot \text{den}(\mathcal{P}^*)$, where $\mathcal{P}^*$ is the optimal solution to the instance (G, D) .*

Lemma 2. *Let $\mathcal{P} = \{P(u, v)\}$ be a partial solution to the instance (G, D) with density $\text{den}(\mathcal{P})$, whose support is a junction tree $J \subseteq G$. For any integer $h \geq 1$, there exists a partial solution $\hat{\mathcal{P}} = \{\hat{P}(u, v)\}$ to the instance $(\hat{G}, D)$ whose support is a junction tree $\hat{J} \subseteq \hat{G}$ with height $\leq h$ and $\text{den}(\hat{\mathcal{P}}) < 8h \sqrt[h]{k/2} \cdot \text{den}(\mathcal{P})$.*

Proof. Recall that J is the union of an in-branching J_I and an out-branching J_O , both rooted at the junction node r . Denote by $\mathcal{D}(\mathcal{P}) \subseteq \mathcal{D}$ the subset of demand pairs connected by the paths in $\mathcal{P}$. Moreover, for each $(u, v) \in \mathcal{D}(\mathcal{P})$, denote by $P(u, r)$ and $P(r, v)$ the segments of $P(u, v)$ contained in J_I and J_O respectively. Observe that node u must be a leaf of J_I and v must be a leaf of J_O , by our assumptions in Section 2.

Now, we shall invoke the height-reduction procedure proposed in [24,17]. We briefly explain its functionality below. Formally, the input must be an integer parameter $h \geq 1$, a branching T , and a non-negative weight function w defined on the edges of T . However, since the output actually does not depend on those edge weights, they need not be specified beforehand. The procedure returns another branching $\bar{T}$ that has height at most h and whose node set $V(\bar{T})$ is a subset of $V(T)$. In particular, $\bar{T}$ has the same root and leaves as T , but may be missing some of the latter's internal nodes. On the other hand, each directed edge $\bar{e} = (u, v)$ of $\bar{T}$ corresponds to the path $P_T(u, v)$ from u to v in T , and is assigned a weight $w(\bar{e}) = \sum_{e \in P_T(u, v)} w(e)$. The salient property of $\bar{T}$ is that

$$\sum_{\bar{e} \in \bar{T}} w(\bar{e}) \leq 2h \sqrt[h]{s/2} \cdot \sum_{e \in T} w(e), \quad (2)$$

where s is the number of leaves of T .

Let $\bar{J}_I$ and $\bar{J}_O$ be the results of applying the aforementioned procedure to J_I and J_O respectively. The junction tree $\bar{J} = \bar{J}_I \cup \bar{J}_O$ is a subgraph of the bi-metric

closure $\bar{G}$ of G . For any pair $(u, v) \in \mathcal{D}(\mathcal{P})$, there exists a unique path $\bar{P}(u, v)$ in $\bar{J}$ that consists of edges corresponding to paths in J which, if concatenated, produce exactly the path $P(u, v)$. Therefore, $\delta(\bar{P}(u, v)) = \delta(P(u, v))$. If $\bar{\mathcal{P}} = \{\bar{P}(u, v) \mid (u, v) \in \mathcal{D}(\mathcal{P})\}$, then using inequality (2) with $\sigma(e)$ substituting for $w(e)$ we derive:

$$\begin{aligned}
c(\bar{\mathcal{P}}) &= \sum_{\bar{e} \in \bar{\mathcal{P}}} \sigma(\bar{e}) + \sum_{\bar{P} \text{ } (u,v) \in \bar{\mathcal{P}}} \delta(\bar{P}(u, v)) \cdot d_{uv} \leq \\
&\leq \sum_{\bar{e} \in \bar{J}_I} \sigma(\bar{e}) + \sum_{\bar{e} \in \bar{J}_O} \sigma(\bar{e}) + \sum_{\bar{P} \text{ } (u,v) \in \bar{\mathcal{P}}} \delta(P(u, v)) \cdot d_{uv} \leq \\
&\leq 2h \sqrt[h]{\frac{|\mathcal{D}(\mathcal{P})|}{2}} \cdot \left(\sum_{e \in J_I} \sigma(e) + \sum_{e \in J_O} \sigma(e) \right) + \sum_{\bar{P} \text{ } (u,v) \in \bar{\mathcal{P}}} \delta(P(u, v)) \cdot d_{uv} \leq \\
&\leq 2h \sqrt[h]{k/2} \cdot \left(2 \sum_{e \in \mathcal{P}} \sigma(e) \right) + \sum_{\bar{P} \text{ } (u,v) \in \bar{\mathcal{P}}} \delta(P(u, v)) \cdot d_{uv} \leq 4h \sqrt[h]{k/2} \cdot c(\mathcal{P}).
\end{aligned}$$

For an edge $\bar{e} \in \bar{J}$, define $\ell(\bar{e})$ as the sum of d_{uv} for pairs $(u, v) \in \mathcal{D}(\mathcal{P})$ such that $\bar{e} \in \bar{P}(u, v)$, and set $\bar{z} = \lfloor \log \ell(\bar{e}) \rfloor$. Among the edges parallel to $\bar{e}$ in $\bar{G}$, let $\hat{e}$ be the one that minimizes $\sigma(\hat{e}) + \delta(\hat{e}) \cdot 2^{\bar{z}}$. Hence, $\hat{e}$ also belongs to the restricted bi-metric closure $\hat{G}$ of G , and furthermore it is easy to see that $\sigma(\hat{e}) + \delta(\hat{e}) \cdot \ell(\bar{e}) < 2(\sigma(\bar{e}) + \delta(\bar{e}) \cdot \ell(\bar{e}))$. We create $\hat{J} \leftarrow \hat{G}$ from $\bar{J}$ (and the corresponding path collection $\hat{\mathcal{P}}$ from $\bar{\mathcal{P}}$) by replacing each edge $\bar{e} \in \bar{J}$ with $\hat{e}$ as above. Clearly, $\hat{\mathcal{P}}$ is a partial solution to the instance $(\hat{G}, D)$ with cost $c(\hat{\mathcal{P}}) < 2 \cdot c(\bar{\mathcal{P}}) \leq 8h \sqrt[h]{k/2} \cdot c(\mathcal{P})$ and density $\text{den}(\hat{\mathcal{P}}) < 8h \sqrt[h]{k/2} \cdot \text{den}(\mathcal{P})$.

Corollary 1. *For any integer $h \geq 1$, there exists a partial solution $\hat{\mathcal{P}}^\circ = \{\hat{P}^\circ(u, v)\}$ to the instance $(\hat{G}, D)$ whose support is a junction tree $\hat{J}^\circ \leftarrow \hat{G}$ with height $\leq h$ and $\text{den}(\hat{\mathcal{P}}^\circ) < \sqrt{k} \cdot 8h \sqrt[h]{k/2} \cdot \text{den}(\mathcal{P}^*) < 8h \cdot k^{1/2+1/h} \cdot \text{den}(\mathcal{P}^*)$.*

4 Approximation Algorithms

$O(k^{1/2+\epsilon})$ approximation. As implied by the discussion in the previous section, the principal task of our algorithm is to determine a low-density partial solution $\hat{\mathcal{P}} = \{\hat{P}(u, v)\}$ to the instance $(\hat{G}, D)$ whose support is a junction tree $\hat{J}$ in $\hat{G}$ with height $\leq h$. (Note that h is an adjustable parameter that is specified in advance of the algorithm's execution.) Then, $\hat{\mathcal{P}}$ is straightforwardly converted into a partial solution $\mathcal{P} = \{P(u, v)\}$ to the original instance (G, D) with at most the same density: for every path $\hat{P}(u, v)$, replace each of its edges with the corresponding path in G and concatenate these paths to form $P(u, v)$. Once we have $\mathcal{P}$, we set $d_{uv} = 0$ for the demand pairs $(u, v) \in \mathcal{D}(\mathcal{P})$ and repeat the same process recursively on the residual instance, until there are no demand pairs left. The final solution is simply the union of all partial solutions produced.

In the following, we describe how to compute $\hat{\mathcal{P}}$. First of all, the algorithm *guesses* the junction node r of the junction tree $\hat{J}^\circ$ in $\hat{G}$ that is implied by

Corollary 1. In practice, this means exhaustively trying out each of n possible choices for r and keeping the one that yields the best result. Assuming we know the correct r , we proceed as follows.

Step 1: Layering and path splitting. We construct an undirected tree $\mathcal{T} = (V(\mathcal{T}), E(\mathcal{T}))$ from $\hat{G}$ by applying the layering and path-splitting procedures presented in Section 3.3 of [8]. We refer the reader to that paper for details of the construction, which are omitted here to save space. Instead, we provide a high-level description of the resulting $\mathcal{T}$ and its main properties.

First of all, every node of $\mathcal{T}$ is a copy of some node of $\hat{G}$. For any $v \in V$, the number of its copies in $V(\mathcal{T})$ is polynomially bounded with respect to the size of $\hat{G}$ (which in turn is polynomial in the size of the problem instance) and exponentially bounded with respect to the parameter h (which does not depend on the instance). Furthermore, each edge $\check{e}$ of $\mathcal{T}$ either (a) joins two distinct copies of the same node $v \in V$, or (b) is an undirected copy of some edge $\hat{e}$ of $\hat{G}$. In the former case, $\check{e}$ is assigned cost coefficients $\sigma(\check{e}) = 0$ and $\delta(\check{e}) = 0$, whereas in the latter we have $\sigma(\check{e}) = \sigma(\hat{e})$, $\delta(\check{e}) = \delta(\hat{e})$, and the endpoints of $\check{e}$ are copies of the endpoints of $\hat{e}$.

Overall, the tree $\mathcal{T}$ is rooted at node r_0 , which is a copy of r , and all its leaves are at distance exactly h from r_0 . For each $v \in V$, denote by $X(v) \subseteq V(\mathcal{T})$ the set of leaves that are copies of v . The construction of $\mathcal{T}$ ensures that for every $(u, v) \in \mathcal{D}$, all paths that connect any node in $X(u)$ with any node in $X(v)$ must pass through the root r_0 . Additionally, consider a collection $\check{\mathcal{P}}$ of paths in $\mathcal{T}$. Among these, let $\check{P}(u, v)$ be the path connecting any node in $X(u)$ with any node in $X(v)$ that minimizes $\delta(\check{P}(u, v))$, if any such path is contained in $\check{\mathcal{P}}$. Setting $\mathcal{D}(\check{\mathcal{P}}) = \{(u, v) \in \mathcal{D} \mid \check{P}(u, v) \text{ exists in } \check{\mathcal{P}}\}$, we define the cost of $\check{\mathcal{P}}$ as

$$c(\check{\mathcal{P}}) = \sum_{\check{e} \in \check{\mathcal{P}}} \sigma(\check{e}) + \sum_{u, v \in \mathcal{D}(\check{\mathcal{P}})} \delta(\check{P}(u, v)) \cdot d_{uv},$$

and its density as $\text{den}(\check{\mathcal{P}}) = c(\check{\mathcal{P}})/|\mathcal{D}(\check{\mathcal{P}})|$, assuming $\mathcal{D}(\check{\mathcal{P}}) \neq \emptyset$. By the construction of $\mathcal{T}$, from any such path collection $\check{\mathcal{P}}$ we may easily derive a partial solution $\hat{\mathcal{P}}$ to the instance $(\hat{G}, D)$ whose support is a junction tree of height $\leq h$ and $\text{den}(\hat{\mathcal{P}}) \leq \text{den}(\check{\mathcal{P}})$. Conversely, for any given partial solution $\hat{\mathcal{P}}$ as above, there exists a collection of paths in $\mathcal{T}$ with density equal to $\text{den}(\hat{\mathcal{P}})$.

Step 2: Bucketing and scaling. We now address the problem of finding a collection $\check{\mathcal{P}}$ of paths in $\mathcal{T}$ with minimum density. A linear programming relaxation of this problem is presented in **LP1**. Each x_e variable indicates whether edge $e \in \check{\mathcal{P}}$; y_{uv} indicates whether there is a path in $\check{\mathcal{P}}$ from any node in $X(u)$ to any node in $X(v)$; and $f_{e,uv}$ indicates whether e is used by such a path. We normalize the sum of all y_{uv} variables to 1, thus the objective function is linear.

LP1 is efficiently solvable, either by first reformulating it into an equivalent linear program with a polynomial number of flow-conservation constraints on the $f_{e,uv}$ variables, or by applying the Ellipsoid method on **LP1** directly, since it admits a polynomial-time separation oracle. Denote by $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{f}^*)$ the resulting optimal fractional solution, whose value OPT is a lower bound on the minimum

$$\begin{aligned}
\mathbf{LP1}: \quad & \min \sum_{e \in E(T)} \left(\sigma(e)x_e + \delta(e) \sum_{u,v \in \mathcal{D}} d_{uv} f_{e,uv} \right) \\
& \text{s.t.} \quad \sum_{u,v \in \mathcal{D}} y_{uv} = 1 \\
& \quad x_e \geq f_{e,uv} \quad \forall e \in E(T), \forall (u,v) \in \mathcal{D} \\
& \quad \sum_{e \in \partial U} f_{e,uv} \geq y_{uv} \quad \forall U \subseteq V(T), \forall (u,v) \in \mathcal{D}: (r_0 \in U) \wedge \\
& \quad \quad \wedge ((U \cap X(u) = \emptyset) \vee (U \cap X(v) = \emptyset)) \\
& \quad x_e - y_{uv} f_{e,uv} \geq 0 \quad \forall e \in E(T), \forall (u,v) \in \mathcal{D}
\end{aligned}$$

density of any collection of paths in $\mathcal{T}$. From the preceding discussion, we deduce that $\text{OPT} \leq \text{den}(\hat{\mathcal{P}}^\circ)$, where $\hat{\mathcal{P}}^\circ$ is the partial solution to the instance $(\hat{G}, D)$ implied by Corollary 1.

Let $q = \lceil \log k \rceil + 1$ and $\mathcal{D}_i = \{(u, v) \in \mathcal{D} \mid 2^{-i} < y_{uv}^* \leq 2^{-i+1}\}$, for $1 \leq i \leq q$. Note that $\sum_{i=1}^q (\sum_{(u,v) \in \mathcal{D}_i} y_{uv}^*) \geq \frac{1}{2}$, therefore at least one of the inner sums must be $\geq 1/(2q)$. If i^* is the index value corresponding to that sum, then $|\mathcal{D}_{i^*}| \geq 2^{i^*-1}/(2q) = 2^{i^*-2}/q$. Hence, we formulate a new linear program **LP2**, by removing the constraint $\sum_{u,v \in \mathcal{D}} y_{uv} = 1$ from **LP1** and fixing the y_{uv} variables as follows: $y_{uv} = 1$ for $(u, v) \in \mathcal{D}_{i^*}$, and $y_{uv} = 0$ for $(u, v) \in \mathcal{D} \setminus \mathcal{D}_{i^*}$.

Step 3: Rounding. Next, we efficiently compute an optimal fractional solution $(\tilde{\mathbf{x}}, \tilde{\mathbf{f}})$ to **LP2**. Its value $\widehat{\text{OPT}}$ is at most 2^{i^*}OPT , because $(\tilde{\mathbf{x}}', \tilde{\mathbf{f}}') = (2^{i^*} \mathbf{x}^*, 2^{i^*} \mathbf{f}^*)$ is a feasible solution to **LP2**.

Consider another feasible solution $(\hat{\mathbf{x}}, \hat{\mathbf{f}}) = (2\tilde{\mathbf{x}}, 2\tilde{\mathbf{f}})$. For each demand pair $(u, v) \in \mathcal{D}_{i^*}$, let $\Delta_{uv} = \sum_{e \in E(T)} \delta(e) \hat{f}_{e,uv} = \frac{1}{2} \sum_{e \in E(T)} \delta(e) \hat{f}_{e,uv}$. The $\hat{f}_{e,uv}$ values represent a fractional 2-unit flow, with sources in $X(u)$ and sinks in $X(v)$. This flow can be decomposed into a collection Π_{uv} of non-disjoint paths, such that each path carries a fraction of the total flow. Observe that all the paths in Π_{uv} pass through r_0 . If $\Pi'_{uv} = \{P \in \Pi_{uv} \mid \delta(P) \leq 2\Delta_{uv}\}$, then by Markov's inequality at least one unit of flow is carried by the paths in Π'_{uv} . We reduce the flow along the paths in $\Pi_{uv} \setminus \Pi'_{uv}$ to zero and, if necessary, we also reduce the flow along paths in Π'_{uv} so that there is exactly one unit of flow remaining in total. Then, we adjust the $\hat{f}_{e,uv}$ values accordingly and prune the $\hat{x}_e$ values so that $\hat{x}_e = \max_{(u,v) \in \mathcal{D}_{i^*}} \{\hat{f}_{e,uv}\}$ for all e . This transformation does not affect the feasibility of $(\hat{\mathbf{x}}, \hat{\mathbf{f}})$.

Finally, we round $(\hat{\mathbf{x}}, \hat{\mathbf{f}})$ to an integral solution $(\bar{\mathbf{x}}, \bar{\mathbf{f}})$ as follows. Consider the edges of $\mathcal{T}$ in breadth-first-search order, starting from the root r_0 . For each edge $e \in E(\mathcal{T})$, either e is incident to r_0 or it has a *parent* edge e' , which is the edge adjacent to e but closer to r_0 . In the first case, set $\bar{x}_e = 1$ with probability $\hat{x}_e$, or $\bar{x}_e = 0$ otherwise. In the second case, if $\bar{x}_{e'} = 0$ then set $\bar{x}_e = 0$; else set $\bar{x}_e = 1$ with probability $\hat{x}_e/\hat{x}_{e'}$, or $\bar{x}_e = 0$ otherwise. Note that $\hat{x}_e \leq \hat{x}_{e'}$, because $\mathcal{T}$ is a tree and all flow paths pass through r_0 . Moreover, denote by $\bar{\mathcal{T}} = \mathcal{T}$

the tree formed by the edges for which $\bar{x}_e = 1$, and let $\bar{\mathcal{D}}_i = \{(u, v) \in \mathcal{D}_i \mid \bar{\mathcal{T}} \text{ contains a path } \check{P}(u, v) \in \Pi'_{uv}\}$. For each $(u, v) \in \bar{\mathcal{D}}_i$, set $\bar{f}_{e,uv} = 1$ along $\check{P}(u, v)$ and $\bar{f}_{e,uv} = 0$ for all other edges. For each $(u, v) \in \mathcal{D}_i \setminus \bar{\mathcal{D}}_i$, set $\bar{f}_{e,uv} = 0$ for all edges. Now, we can efficiently convert $\check{\mathcal{P}} = \{\check{P}(u, v) \mid (u, v) \in \bar{\mathcal{D}}_i\}$ into a partial solution $\hat{\mathcal{P}}$ to the instance $(\hat{G}, D)$, with $\text{den}(\hat{\mathcal{P}}) \leq \text{den}(\check{\mathcal{P}})$.

Analysis. It is straightforward to realize that the expected cost of $\check{\mathcal{P}}$ is at most the expected value of $(\bar{\mathbf{x}}, \bar{\mathbf{f}})$, which does not exceed $\sum_{e \in E(\mathcal{T})} (\sigma(e)\bar{x}_e + \delta(e) \sum_{(u,v) \in \mathcal{D}_i} d_{uv} \bar{f}_{e,uv}) \leq 2 \cdot \widetilde{\text{OPT}}$. For each $(u, v) \in \mathcal{D}_i$, with probability $\Omega(h^{-1})$ there exists a path in $\bar{\mathcal{T}}$ from r_0 to any node of $X(u)$ that is an end-point of a path in Π'_{uv} . This is deduced from the arguments in the proof of Theorem 3.4 in [14], adapted to the special case of a tree with height $\leq h$. The same is true for $X(v)$, by symmetry, and the two events are independent because any node in $C(u_0)$ and any node in $C(v_{2h})$ have no common ancestor in $\mathcal{T}$ except r_0 . Hence, the probability that at least one path in Π'_{uv} is contained in $\bar{\mathcal{P}}$ is $\Omega(h^{-2})$, which is a constant. As a result, with high probability $|\bar{\mathcal{D}}_i| = \Omega(|\mathcal{D}_i|)$. Alternatively, we may perform the rounding using the deterministic techniques from [6], so that an analogous property holds with probability 1. Therefore $\text{den}(\hat{\mathcal{P}}) \leq O(\widetilde{\text{OPT}}/|\mathcal{D}_i|) \leq O(2^i \text{OPT}/(2^{i-2}/q)) \leq O(\log k) \cdot \text{OPT}$, and thus $\text{den}(\hat{\mathcal{P}}) \leq O(\log k) \cdot \text{OPT} \leq O(\log k) \cdot \text{den}(\hat{\mathcal{P}}^\circ)$. Consequently, we have proven the following:

Lemma 3. *The algorithm computes a partial solution $\hat{\mathcal{P}}$ to the instance $(\hat{G}, D)$ with density $\text{den}(\hat{\mathcal{P}}) \leq O(k^{1/2+1/h} \log k) \cdot \text{den}(\mathcal{P}^*)$, where $\mathcal{P}^*$ is the optimal solution to the instance (G, D) .*

As mentioned earlier, $\hat{\mathcal{P}}$ is easily transformed into a partial solution $\mathcal{P}$ to the instance (G, D) with $\text{den}(\mathcal{P}) \leq \text{den}(\hat{\mathcal{P}})$. By applying a well-known argument for recursive greedy covering (see e.g. [5,13]), we establish that the final solution produced by our algorithm has density at most $O(k^{1/2+1/h} \log k) \cdot \text{den}(\mathcal{P}^*)$. Additionally, the overall running time is polynomial in the size of the instance. Setting the parameter h appropriately, e.g. $h = \lceil 2/\epsilon \rceil$, guarantees that:

Theorem 1. *For any constant $\epsilon > 0$, the directed buy-at-bulk network design problem may be efficiently approximated within an $O(k^{1/2+\epsilon})$ factor.*

$O(n^{4/5+\epsilon})$ approximation. We start by guessing the value of $c(\mathcal{P}^*)$, within a factor of 2 (so that we need to make only a polynomial number of guesses). Furthermore, for each demand pair (u, v) , define a node set $B(u, v)$ as follows. The node $w \in V$ belongs to $B(u, v)$ if and only if there exist in G two paths P_1 and P_2 , from u to w and from w to v respectively, such that (a) $\sigma(P_1), \sigma(P_2) \leq c(\mathcal{P}^*)/n^{4/5}$, and (b) $\delta(P_1), \delta(P_2) \leq n^{4/5} \cdot c(\mathcal{P}^*)/k$. We say that a demand pair (u, v) is *good* if $|B(u, v)| \geq n^{2/5}$, and *bad* otherwise. Denote by $\mathcal{D}_g$ and $\mathcal{D}_b$ the sets of good and bad pairs respectively, with $k_g = |\mathcal{D}_g|$ and $k_b = |\mathcal{D}_b|$. We have:

Lemma 4. *For any $\tau \in \{g, b\}$, we can efficiently compute a partial solution $\mathcal{P}_\tau$ such that $\mathcal{D}(\mathcal{P}_\tau) = \mathcal{D}_\tau$ and $c(\mathcal{P}_\tau) \leq O(n^{4/5+\epsilon}) \cdot c(\mathcal{P}^*)$.*

This lemma, whose proof is omitted here due to space constraints, leads immediately to our second main result.

Theorem 2. *For any constant $\epsilon > 0$, the directed buy-at-bulk network design problem may be efficiently approximated within an $O(n^{4/5+\epsilon})$ factor.*

Approximations for the single-sink case. If all demand pairs share a common sink (or source, symmetrically), then a simple argument enables us to assume that the support of the optimal solution is a junction tree. This signifies an improvement of the quantitative claim in Lemma 1 by a factor of $\sqrt{k}$, and said improvement propagates to Theorem 1, yielding the corollary below.

Corollary 2. *For any constant $\epsilon > 0$, the single-sink directed buy-at-bulk network design problem may be efficiently approximated within an $O(k^\epsilon)$ factor.*

The above corollary also implies an $O(n^\epsilon)$ approximation for the single-sink case, because in the latter the number k of demand pairs cannot exceed $n - 1$.

5 Concluding Remarks and Open Problems

Recall that, in undirected graphs, buy-at-bulk network design is much harder to approximate than Steiner forest—and provably so, modulo a reasonable complexity assumption. By contrast, as a consequence of this work, the current picture in the directed case appears to be quite the opposite, i.e. the two problems have essentially identical approximation guarantees. A potential explanation may be that directionality is now the dominant factor determining problem hardness, overshadowing the effects of the cost model. On the other hand, our results could also be construed as a small piece of evidence that there exist, in fact, better approximation algorithms for directed Steiner forest, thus meriting further investigation. Last but not least, by the same token it may be possible to obtain a tighter inapproximability bound for directed buy-at-bulk network design than for directed Steiner forest, e.g. of the form $\Omega(n^\lambda)$ for some positive λ .

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